

Day 18 — Asynchronous Programming in Node.js

Challenge 7: Callbacks

Callbacks.js

```
const fs = require("fs");

fs.readFile("data.txt", "utf8", (err, data) => {
  if (err) {
    console.log("Error reading file:", err);
    return;
  }

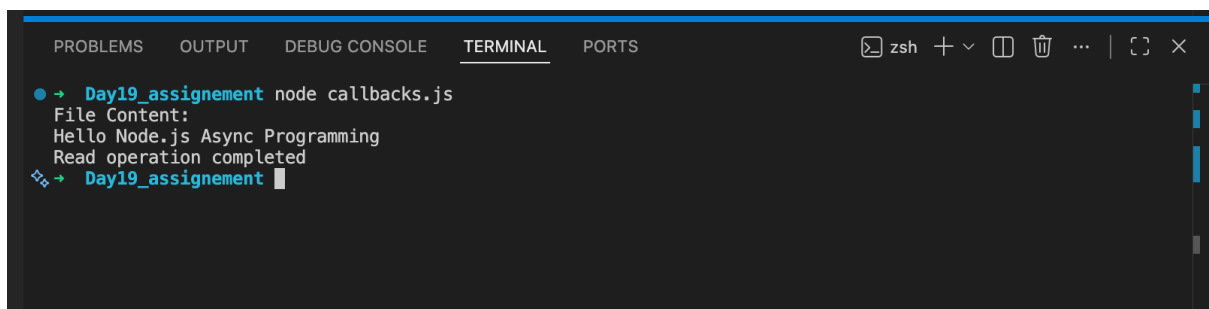
  console.log("File Content:");
  console.log(data);

  setTimeout(() => {
    console.log("Read operation completed");
  }, 1000);
});
```

Data.txt

Hello Node.js Async Programming

Output:

A screenshot of a terminal window with a dark background. The terminal has tabs at the top: PROBLEMS, OUTPUT, DEBUG CONSOLE, TERMINAL (selected), and PORTS. The output shows the execution of 'node callbacks.js' from the 'Day19_assignment' directory. The output lines are: 'File Content:', 'Hello Node.js Async Programming', and 'Read operation completed'. There is a small delay between the second and third line. The prompt 'Day19_assignment' is visible at the bottom.

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
● → Day19_assignment node callbacks.js
File Content:
Hello Node.js Async Programming
Read operation completed
❖ → Day19_assignment
```

Challenge 8: Promises

Promises.js

```
const fs = require("fs").promises;

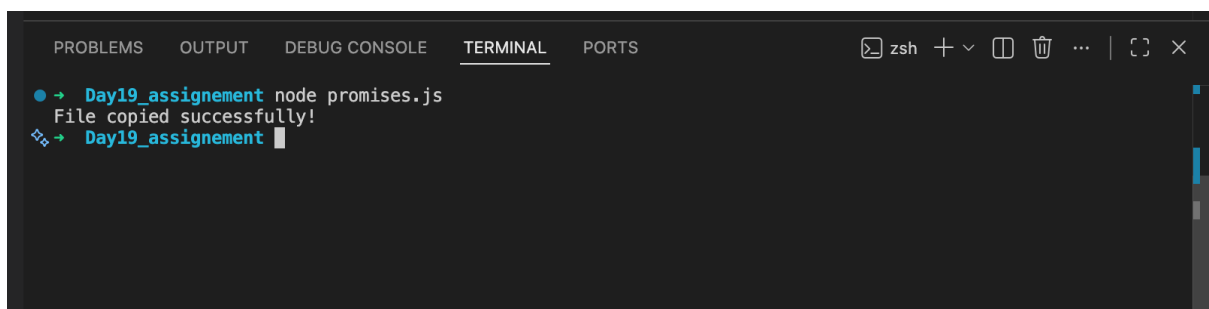
fs.readFile("input.txt", "utf8")
  .then(data => {
    return fs.writeFile("output.txt", data);
  })
  .then(() => {
    console.log("File copied successfully!");
  })
  .catch(err => {
    console.log("Error:", err);
  });
```

input.txt

This file will be copied using Promises

Generated output.txt file: This file will be copied using Promises

Output:

A screenshot of a terminal window with a dark background. The terminal has tabs at the top: 'PROBLEMS', 'OUTPUT', 'DEBUG CONSOLE', 'TERMINAL' (which is active and underlined), and 'PORTS'. On the right side of the terminal, there are icons for a shell (zsh), a plus sign, a dropdown arrow, a window icon, a trash can, and a close button. The terminal output shows a blue prompt character followed by 'Day19_assignment' and the command 'node promises.js'. The output of the command is 'File copied successfully!'. Below this, there is another blue prompt character followed by 'Day19_assignment' and a cursor.

Challenge 9: Async/Await

asyncAwait.js:

```
const fs = require("fs").promises;

async function copyFile() {
  try {
    const data = await fs.readFile("input.txt", "utf8");

    await new Promise(res => setTimeout(res, 1000));

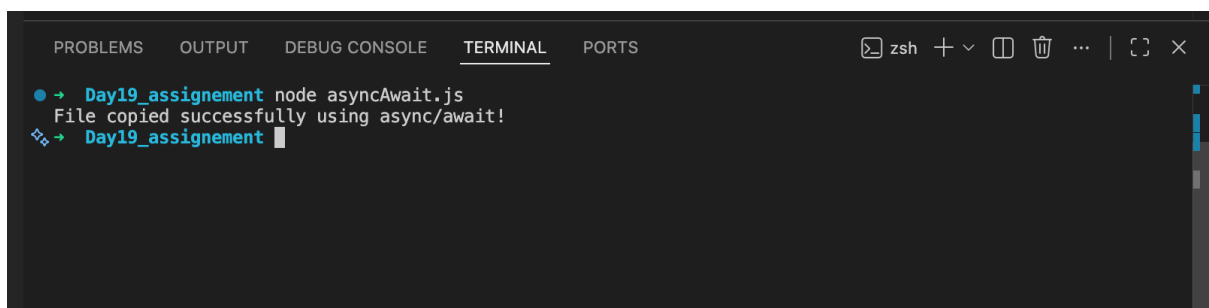
    await fs.writeFile("output2.txt", data);

    console.log("File copied successfully using async/await!");
  } catch (err) {
    console.log("Error:", err);
  }
}

copyFile();
```

Generated output2.txt file: This file will be copied using Promises

Output:



Guide to run the file:

To run the the callbacks.js enter : node callbacks.js command in the terminal

To run the the promises.js enter : node promises.js command in the terminal

To run the the asyncAwait.js enter : node asyncAwait.js command in the terminal